

LESSON 328 (3.5 DUAL XC, 3.3 INSTRUMENT)

Lesson Objective

During this lesson, the student will become more familiar with ATC departures, enroute, and arrival procedures. This cross-country must be at least 250 nautical miles on Federal Airways or as routed by ATC. The student must complete three different instrument approaches at different airports or otherwise comply with **14 CFR Part 141 Appendix C, 4 (c)(1)(ii).**

(ii) Is a distance of at least 250 nautical miles along airways or ATC-directed routing with one segment of the flight consisting of at least a straight-line distance of 100 nautical miles between airports;

(iii) Involves an instrument approach at each airport; and

(iv) Involves three different kinds of approaches with the use of navigation systems.

Briefing

- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Review homework assigned in the previous lesson.
- Student questions.

Lesson Content (Ground)

Go/no-go Decision

- The student must make a go/no-go decision depending on:
 - Personal minimums.
 - FIT minimums.
 - Aircraft.
 - Airspace.
- Review the students planning with a focus on:
 - Weather, weight and balance, performance, flight planning, and fuel calculations.
 - NOTAMs, TFRs, FIFs, aircraft documents, and personal documents.
 - Equipment requirements (e.g. flashlight, fuel card, drinking water, view-limiting devices, life jackets, fly away kit, survival kit, etc.)
 - FRAT/TEM/safety considerations for the flight.
- While this is represented by one line item prior to departure, the go/ no-go decision is continuous and should be a consideration even once airborne.

Lesson Content (Flight)

Departure Procedures and Clearances

- **Use of DPs and/or radar**
 - Review any published SIDs/ODPs and consider aircraft performance to ensure they can be complied with.
 - KSFB - Sanford usually assigns the Sanford One Departure, a simple radar vector-based SID. This should be briefed on the ground before the flight.

- Make sure the student correctly inputs assigned transponder codes as well as understands the ALT and IDENT functions.

Enroute Procedures and Clearances

- **VOR, and/or GPS navigation**
 - Ensure the student is proficient with navigation using the VOR and/or GPS. Make sure they can identify and track airways using both the VOR and the GPS, as well as navigate to GPS fixes.
 - Brief the student on the use of the FPL page.
 - Emphasize the importance of proper setup prior to taxi and takeoff; the flight plan should be loaded in the GPS, the departure heading and altitude should be bugged on the PFD, and the departure frequency and transponder code should be set *prior* to entering the movement area.

Arrival Procedures and Clearances

- **Use of STARs and/or radar vectors**
 - Ensure the student complies with any assigned STARs or radar vectors issued by ATC.

Instrument approach #1

Instrument approach #2

Instrument approach #3

Circling approach

- On any one of the approaches, request a circle to land.
 - If this cannot be completed at KMLB, it can be completed at X26 without talking to ATC. Make sure to make calls to CTAF and circle at pattern altitude if needed.
- Remind the student of the circling radius airspace.
 - The  symbol indicates expanded circling radius. A lack of this symbol indicates standard circling radius.
- Just prior to the specified distance, the student should go visual without being told, and begin the circle.
 - During their approach briefing, the student should include the direction (left/right) that they will turn when they begin the circle so that they don't have to figure that out now.
- Inform ATC if required.
- Hold the circling MDA throughout the circle.
- Once a landing is assured, descend below the circling MDA, extend the flaps and land.

Post-Brief

- Student Self Critique.
- Instructor critique.
- Questions from the student.
- Lesson grading.
- Logbook completion.

- Schedule the next lesson.
- Preview the next lesson → 329
- Assign Homework:
 - Review all Stage 2 ground knowledge in preparation for lesson 329.

Completion Standards

At the completion of this flight, the student shall:

1. Have a thorough understanding of cross-country and emergency operations.

This lesson must comply with the regulatory requirements of 14 CFR Part 141, Appendix C, 4 (c) (1) (ii).

(ii) Is a distance of at least 250 nautical miles along airways or ATC-directed routing with one segment of the flight consisting of at least a straight-line distance of 100 nautical miles between airports;

(iii) Involves an instrument approach at each airport; and

(iv) Involves three different kinds of approaches with the use of navigation systems.